

Transformers: 2017 vs 2026

(Multimodal LLMs, MoE, Attention at Scale)

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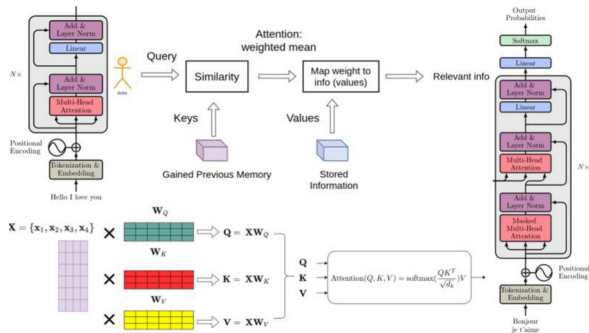
June 12, 2026

1. Roadmap: 2017 to 2026 — four key engineering changes
2. Baseline (2017): encoder–decoder Transformer, self-attention, and multi-head attention
3. Positional information: sinusoidal encodings, RoPE, and ALiBi
4. Normalization: transformer block, Post-LN $\rightarrow$ Pre-LN, and RMSNorm
5. Inference memory and attention variants: KV cache, MHA/MQA/GQA, and decoder-only LLMs
6. Mixture of Experts (MoE): routing, milestones, and Mixtral
7. Multimodality: vision–language models and fusion patterns (early, middle, late)
8. Synthesis and discussion: 2017 vs. 2026 comparison and summary
9. References and further reading

Four Key Engineering Changes (2017 → 2026)

1. **Bandwidth matters more than just speed:** New attention methods like FlashAttention focus on moving data efficiently and using memory well (built-in memory or RAG based retrieval).
2. **From fixed to flexible compute:** Models now adjust their computation and memory use based on the input (using tools like RoPE, sparse MoE, and adaptive resolution).
3. **From separate to unified input streams:** Instead of having different pieces for text, images, etc., everything is processed together in one sequence—with new practical challenges.
4. **More thinking during use:** Some new models use extra internal steps ("thinking tokens") for harder questions, shifting cost to longer decoding instead of just bigger models.

The 2017 Transformer



- ▶ The original 2017 Transformer was designed to address key limitations of Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) units:
 - Lack of parallelization
 - Vanishing gradient problem inherent in sequential processing
- ▶ Replaced recurrence with self-attention to allow all tokens in a sequence to be processed simultaneously during training¹
- ▶ Enabled significant reductions in wall-clock time needed for model convergence

¹See Vaswani et al., “Attention is All You Need,” 2017.

► What is a Transformer?

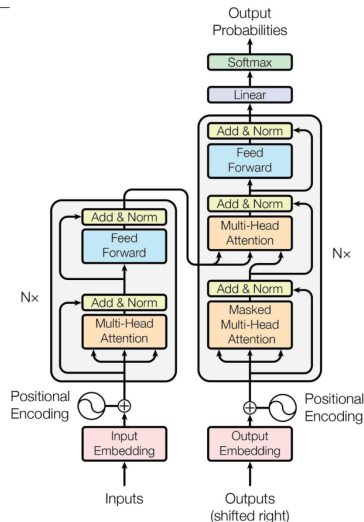
- Introduced by Vaswani et al. (2017) – “Attention is All You Need”

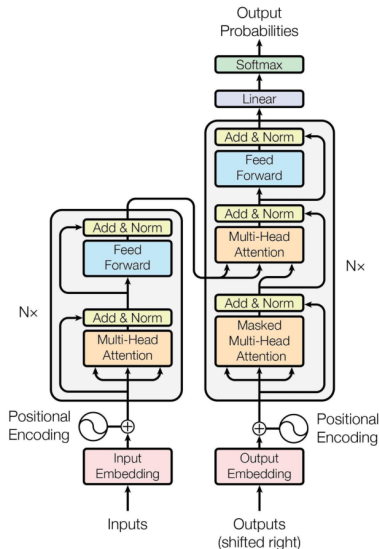
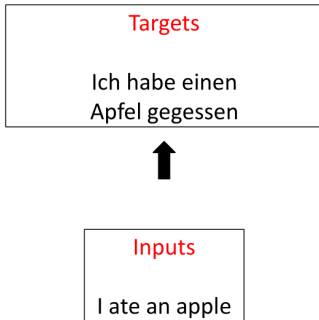
► Key features:

- No recurrence, all attention
- Encoder-decoder structure
- Enables parallel processing
- Scales well with data and compute

Transformer Architecture: Components Overview

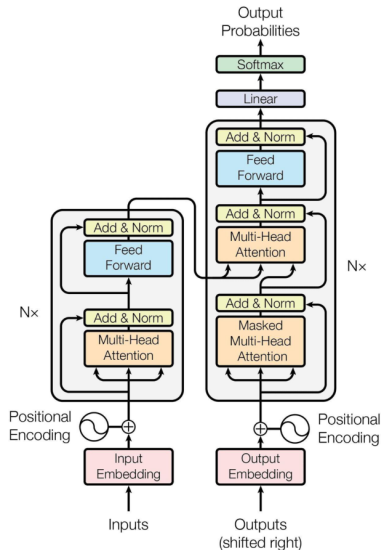
- ▶ Tokenization
- ▶ Input Embeddings
- ▶ Position Encodings
- ▶ Query, Key, & Value
- ▶ Attention
- ▶ Self Attention
- ▶ Multi-Head Attention
- ▶ Feed Forward
- ▶ Add & Norm
- ▶ Encoders
- ▶ Masked Attention
- ▶ Encoder Decoder Attention
- ▶ Linear
- ▶ Softmax
- ▶ Decoders
- ▶ Encoder-Decoder Models





Processing Inputs

Inputs
I ate an apple



- ▶ Computes contextual representation of a word using all other words
- ▶ **Inputs:** Query (Q), Key (K), Value (V)
- ▶ **Output:** Weighted sum of values

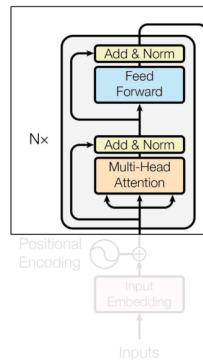
Attention formula:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

- ▶ Enables each word to attend to all others

- ▶ **Query:** What we're looking for (e.g., the meaning of “bank” in context)
- ▶ **Keys:** Tags or features associated with each word/sentence (e.g., “river”, “money”)
- ▶ **Values:** Information linked to each key (e.g., context-specific meaning)
- ▶ Uses dot-product similarity between Query and Keys to assign attention weights
- ▶ Weighted sum of Values produces the contextual representation

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



The

animal

didn't

cross

the

street

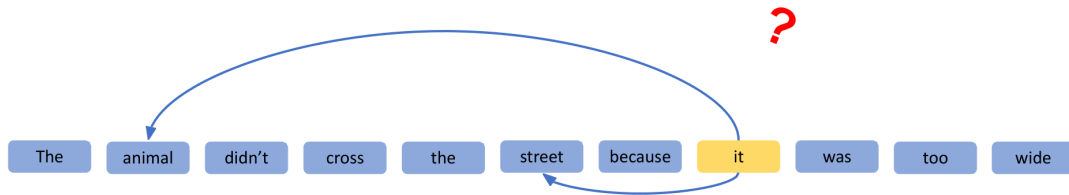
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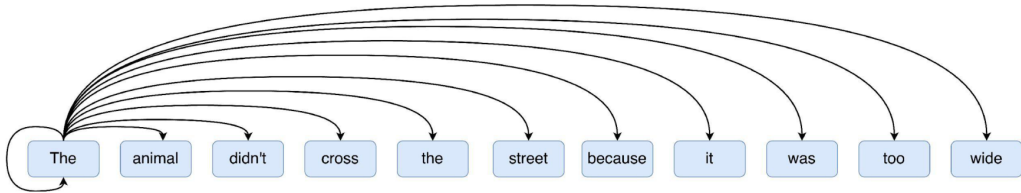
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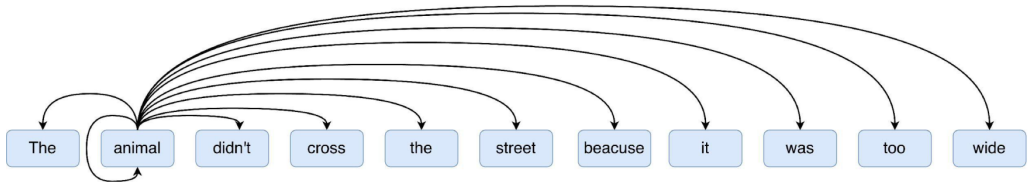
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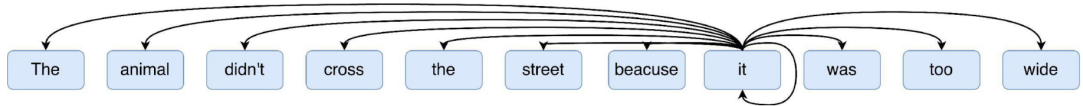
wide

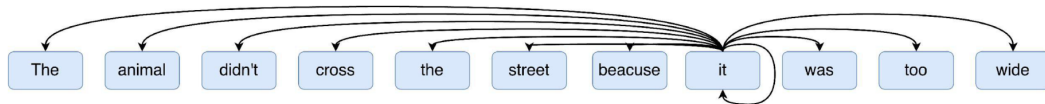


coreference resolution?









SELF

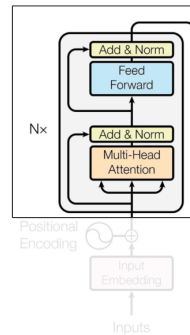
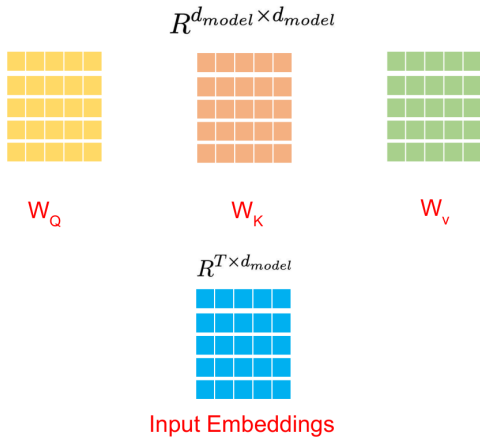
Query Inputs

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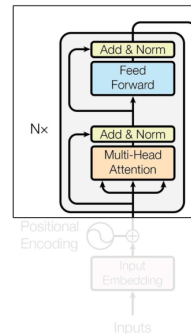
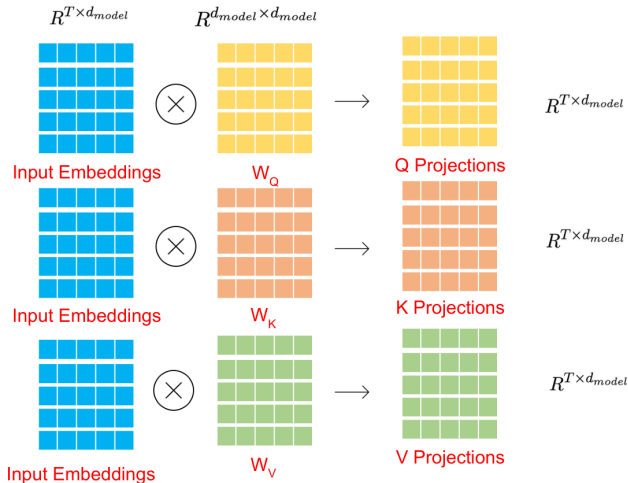
Key Inputs

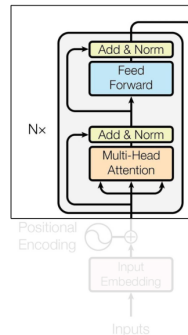
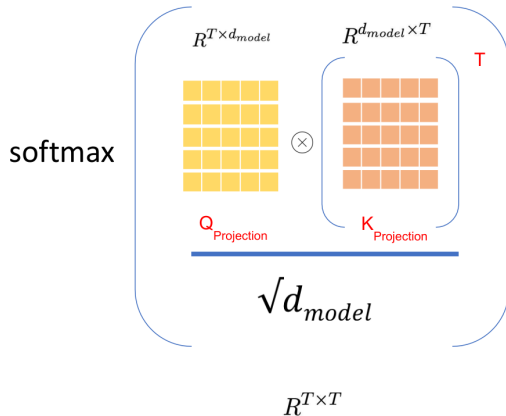
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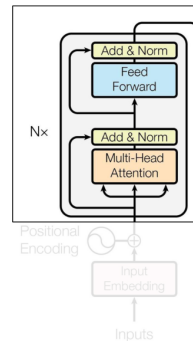
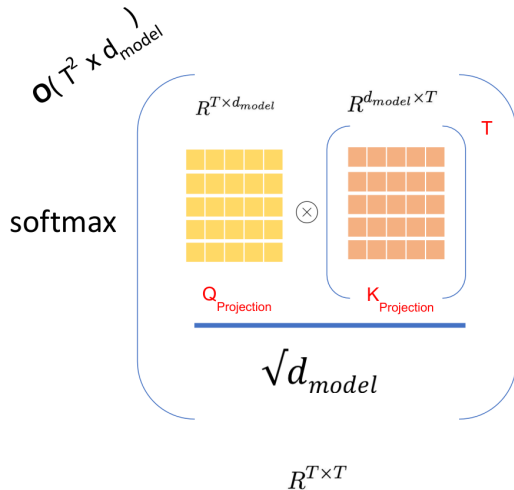
Value Inputs

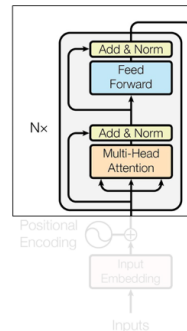
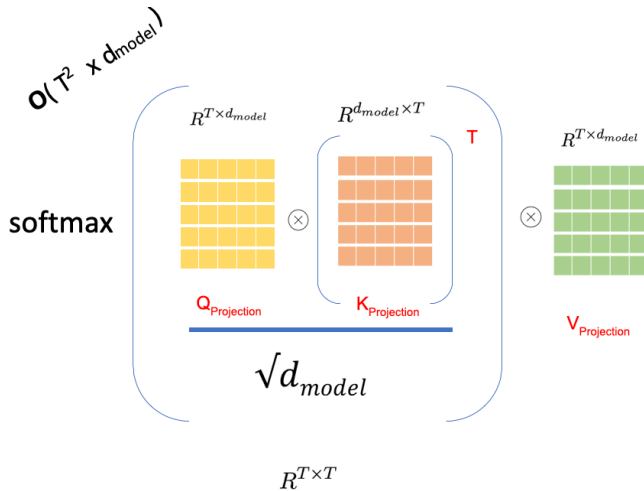


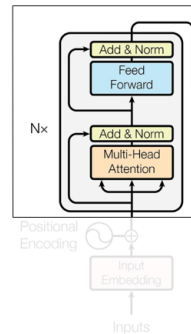
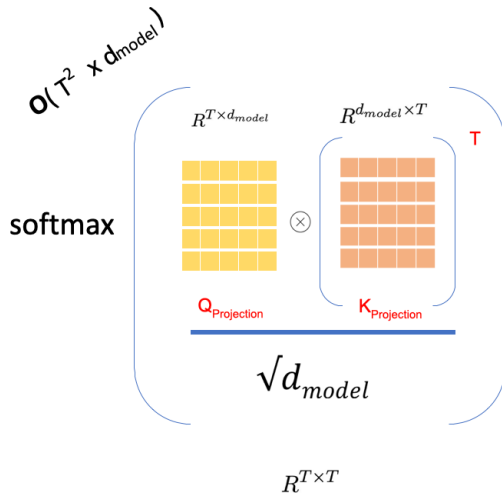
Self Attention

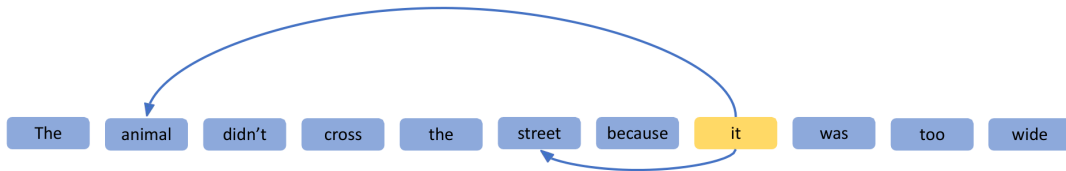




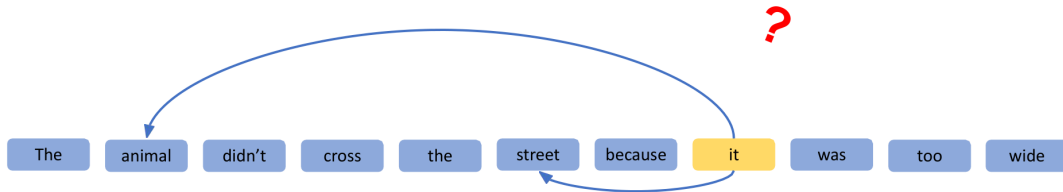








Coreference resolution ✓



Sentence boundaries ?

Coreference resolution

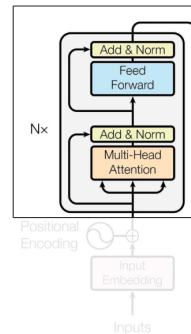
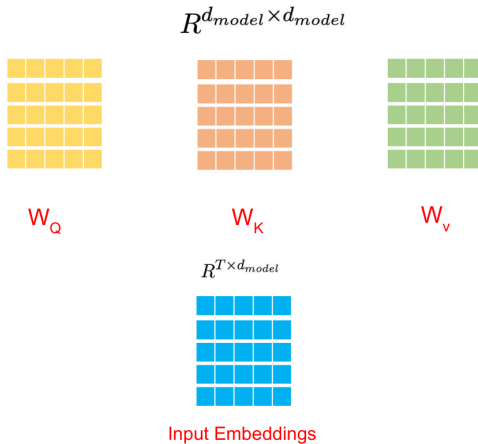


Context ?

Semantic relationships ?

Part of Speech ?

Comparisons ?



- ▶ Multi-head attention allows the model to jointly attend to information from different representation subspaces at different positions.
- ▶ It consists of multiple attention heads, each with its own set of parameters.
- ▶ The outputs of all heads are concatenated and linearly transformed.

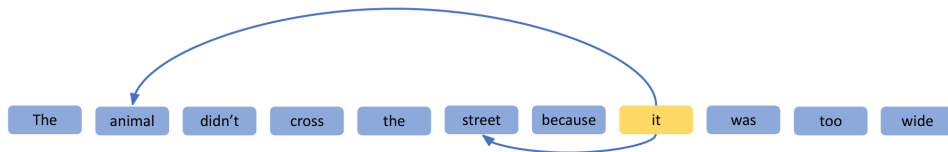
Formula:

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$$

where each head is defined as:

$$\text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$

- ▶ W_i^Q , W_i^K , and W_i^V are learned projection matrices for each head.
- ▶ W^O is the output projection matrix that combines the outputs of all heads.



Sentence boundaries ✓

Coreference resolution ✓

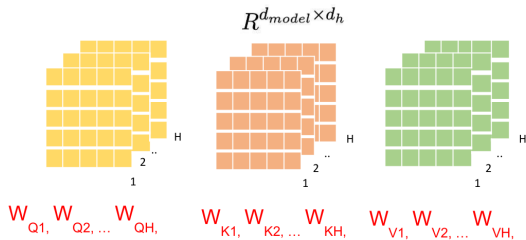
Context ✓

Semantic relationships ✓

Part of speech ✓

Comparisons ✓

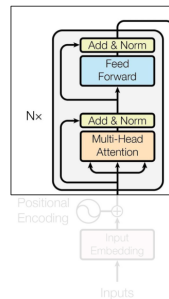
Multi-Head Attention



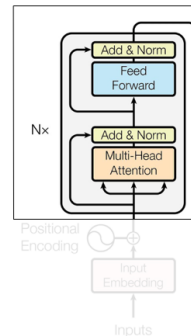
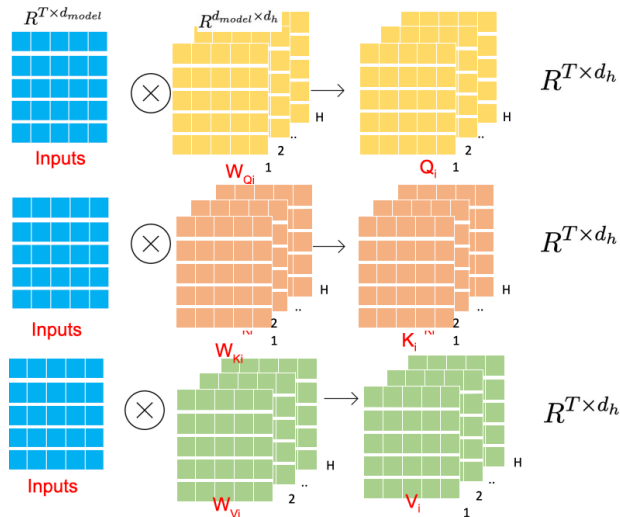
Input Embeddings

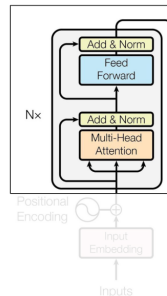
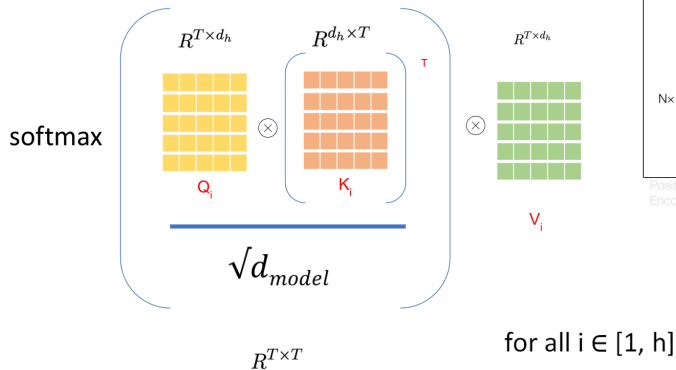
$$R^{T \times d_{model}}$$

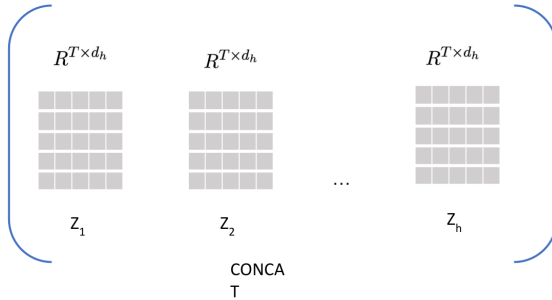
$$d_h = \frac{d_{model}}{h}$$



Multi-Head Attention



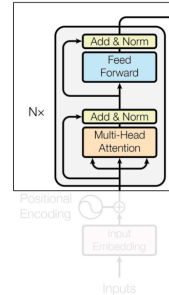


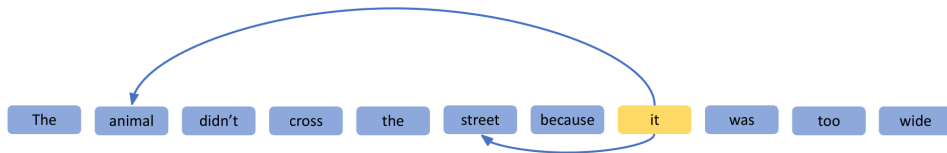


Multi Head Attention : Z

$$d_h = \frac{d_{model}}{h}$$

$$R^{T \times d_{model}}$$





Sentence boundaries



Coreference resolution



Context



Semantic relationships



Part of speech



Comparisons



- ▶ Enables the model to jointly attend to information from different representation subspaces at different positions.
- ▶ Captures various linguistic and syntactic patterns by using multiple attention heads.
- ▶ Improves the model's ability to represent complex relationships in the data.
- ▶ Provides richer and more diverse feature representations.

- ▶ Transformers lack sequence order awareness.
- ▶ **Positional encodings** are added to input embeddings to provide information about token positions.
- ▶ **Sinusoidal encoding:**

$$PE(pos, 2i) = \sin\left(\frac{pos}{10000^{\frac{2i}{d_{\text{model}}}}}\right)$$
$$PE(pos, 2i + 1) = \cos\left(\frac{pos}{10000^{\frac{2i}{d_{\text{model}}}}}\right)$$

For each position, an embedded input is moved the same distance but at a different angle. **Inputs that are close to each other in the sequence have similar perturbations, but inputs that are far apart are perturbed in different directions.**

$$PE_{(pos, 2i)} = \sin \left(pos / 10000^{2i/d_{model}} \right)$$

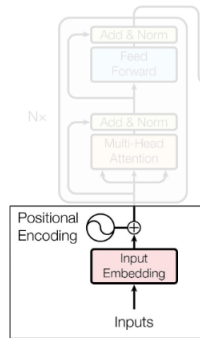
$$PE_{(pos, 2i+1)} = \cos \left(pos / 10000^{2i/d_{model}} \right)$$

pos idx of the token in input sentence

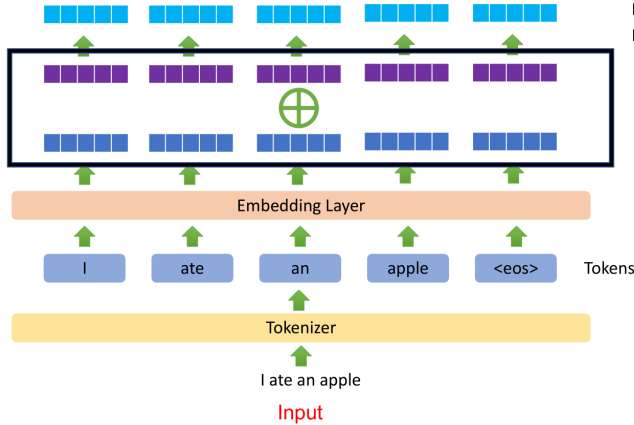
i i^{th} dimension out of d

d_{model} embedding dimension of each token

Different calculations for odd and even embedding indices.



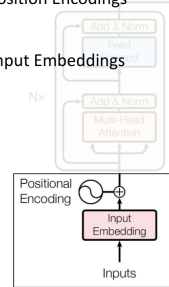
Position Encoding



Final Input Embeddings

Position Encodings

Input Embeddings



▶ Learnable position embeddings:

- Position embeddings are parameters learned during training.
- Allow the model to adapt position representations to the task.

▶ Rotary Position Embeddings (RoPE):

- Encodes positions by rotating query and key vectors in multi-head attention.
- Enables better extrapolation to longer sequences.

▶ ALiBi (Attention with Linear Biases):

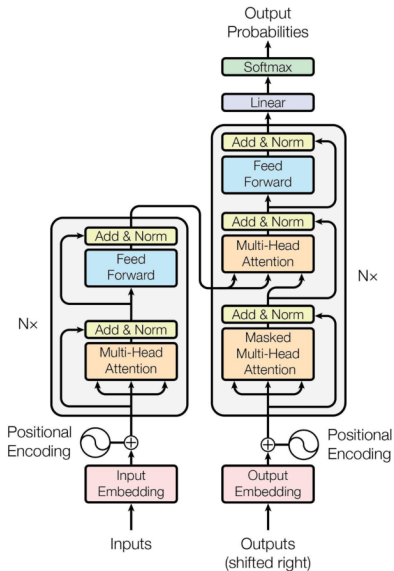
- Adds linear biases to attention scores based on distance between tokens.
- Encourages attention to nearby tokens without explicit position embeddings.

- ▶ **Multi-head Attention**
- ▶ **Add & Norm**
- ▶ **Feed-forward Layer**
- ▶ **Add & Norm**

Note

Residual connections and layer normalization are crucial for effective training.

Transformer Block



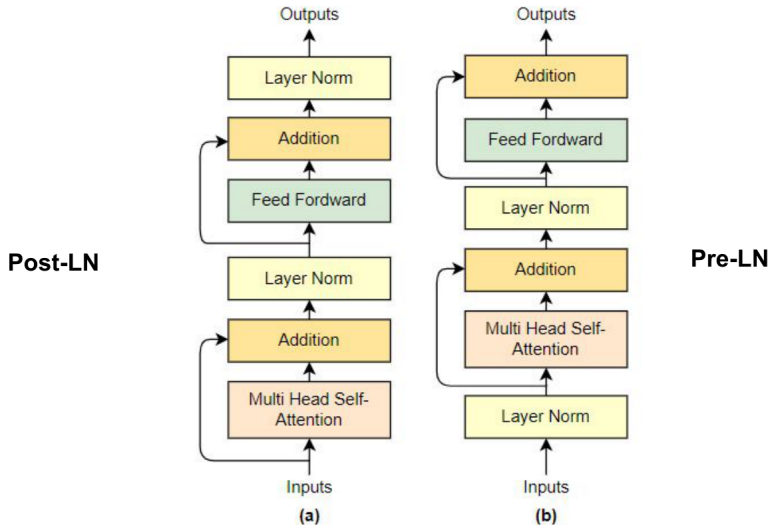
Feature	Pre-LN	Post-LN
Stability	✓ More stable	✗ Less stable
Gradient flow	✓ Better	✗ Can explode/vanish
Used in	GPT-3, T5	Original Transformer

Table 1: Comparison of Pre-LayerNorm and Post-LayerNorm Architectures

Pre-LN: LayerNorm applied **before** sublayers

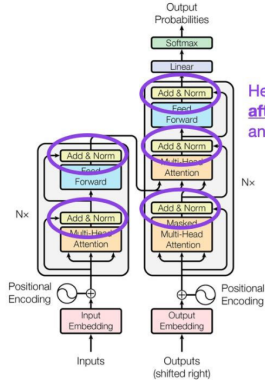
Post-LN: LayerNorm applied **after** sublayers

Pre-LayerNorm vs. Post-LayerNorm



Pre-LayerNorm vs. Post-LayerNorm

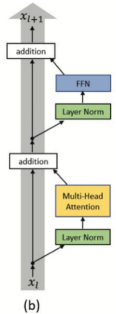
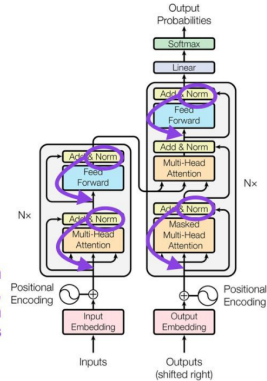
Post-LN Transformer



Here, LayerNorms are after attention and after fully connected layers

Better gradients if LayerNorm is placed inside residual connection, before the attention and fully connected layers

Pre-LN Transformer



Places the layer normalization between the residual blocks: the expected gradients of the parameters near the output layer are large

LayerNorm: Normalizes across the feature dimension by subtracting the mean and dividing by the standard deviation.

RMSNorm: Removes the mean and scales by the root mean square (RMS) of the features.

- ▶ Simpler
- ▶ Faster
- ▶ Slightly better in large-scale settings.

- ▶ In **decode**, the model emits **one token per step**, but that token still depends on the **key and value** tensors for **all** earlier positions.
- ▶ Those positions include the **prompt** (K/V from **prefill**) plus any **new** K/V computed up to the current step.
- ▶ **KV caching** keeps these tensors in GPU memory so they are **not recomputed** at every step; each iteration **appends** the newly computed K/V to the cache for the next step.
- ▶ In many stacks there is **one KV cache per transformer layer**.

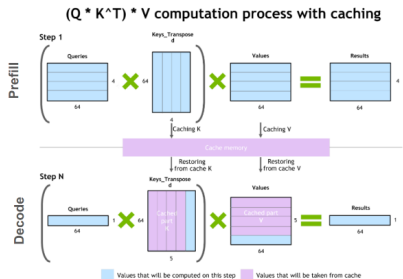


Figure 1: illustration of the key-value caching mechanism.

Verma & Vaidya, "Mastering LLM Techniques: Inference Optimization," NVIDIA Technical Blog (2023). [Article](#) — Key-value caching (Figure 1).

- ▶ Major GPU memory uses for serving: **model weights** and the **KV cache** (self-attention tensors cached to skip redundant work).
- ▶ Under batching, each request still needs a **separate** KV-cache allocation—often the limiting factor before weights alone.
- ▶ Cost scales **linearly** with batch and length—caps throughput and long-context use; motivates MQA/GQA, paging (PagedAttention), KV quantisation, etc.

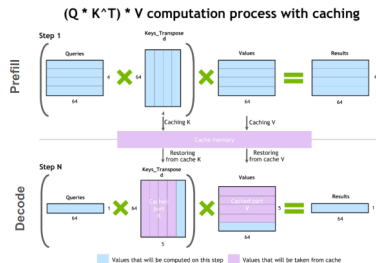
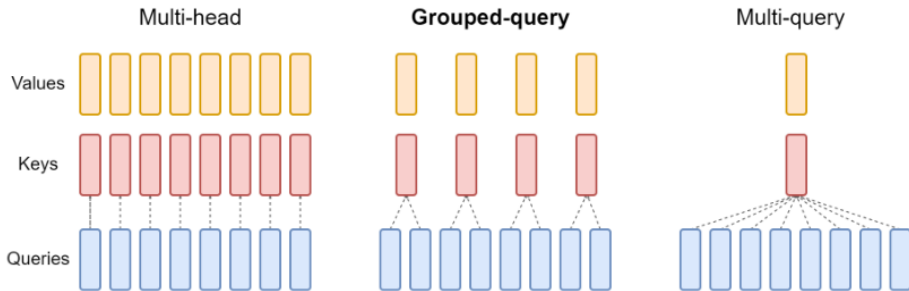


Figure 1: illustration of the key-value caching mechanism (from previous frame).

MHA, MQA, GQA: shrinking the KV cache



- **Multi-head attention (MHA):** H separate query, key, and value heads.
- **Multi-query attention (MQA):** single shared key and value heads for all query heads.
- **Grouped-query attention (GQA):** shares key and value heads for each *group* of query heads.
- GQA conceptually interpolates between multi-head and multi-query attention.

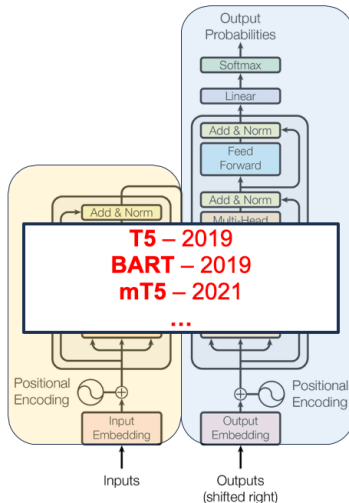
Reference: Ainslie et al., "GQA: Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints,"

The LLM Era – Paradigm Shift in Machine Learning

BERT – 2018
DistilBERT – 2019
RoBERTa – 2019
ALBERT – 2019
ELECTRA – 2020
DeBERTa – 2020

...

Representation



GPT – 2018
GPT-2 – 2019
GPT-3 – 2020
GPT-Neo – 2021
GPT-3.5 (ChatGPT) – 2022
LLaMA – 2023
GPT-4 – 2023

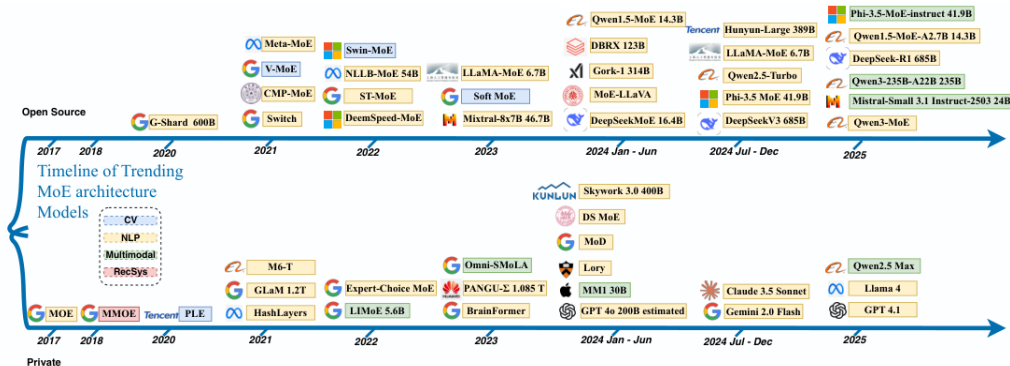
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Generation

- ▶ GPT-style **causal (masked) self-attention**: each token attends only to the past.
- ▶ Single stack replaces separate encoder/decoder for many language (and vision–language) workloads.
- ▶ Encoder–decoder Transformers remain standard in **seq2seq** (MT, Whisper ASR, some VL captioning) where cross-attention helps.

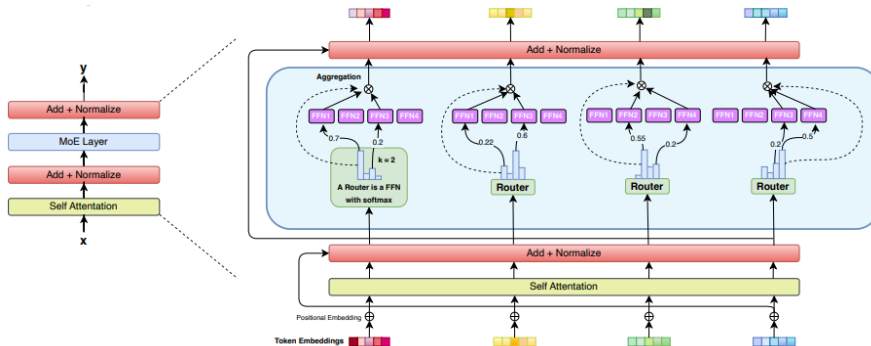
- ▶ A dense feed-forward block always runs the same large MLP for every token. An MoE block replaces that MLP with several smaller expert MLPs.
- ▶ A lightweight router (gating network) looks at each token and picks one or a few experts to run. The rest stay idle for that token.
- ▶ Why it helps: you can increase total parameters (model capacity) without increasing active FLOPs per token as much as a fully dense model—similar spirit to “widen capacity, keep per-token work bounded.”
- ▶ Routing can become unbalanced, communication-heavy in distributed training, or collapse onto a few experts unless training is regularized.

Sources: Zhang et al., “Mixture of Experts in Large Language Models,” [arXiv:2507.11181](#); Kranen & Nguyen, [NVIDIA Technical Blog](#) (Mar. 2024).



Reference: Zhang et al., "Mixture of Experts in Large Language Models," [arXiv:2507.11181](https://arxiv.org/abs/2507.11181).

MoE for a decoder-only Transformer



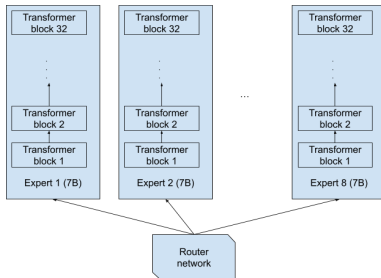
MoE for a

decoder-only Transformer with $k = 2$ experts.

- ▶ Input x enters a **gating / router** that outputs scores for each expert.
- ▶ Only the top- k experts run their FFN; outputs are **combined** (often a weighted sum) for the residual stream.

Example: Mixtral 8×7B — a common misconception

- ▶ The name sounds like “eight separate 7B models.” The **released architecture is not that**: experts are **not** eight full standalone stacks.
- ▶ Instead, think “**eight expert FFNs** per MoE layer, with **shared** attention and norms,” and only a subset fires per token (here, **top-2** experts).



Reference: Kranen & Nguyen, “Applying Mixture of Experts in LLM Architectures,” NVIDIA Technical Blog (2024), [article](#).

MoE: Most Preferred Tokens By Experts



Reference: Kranen & Nguyen, “Applying Mixture of Experts in LLM Architectures,” NVIDIA Technical Blog (2024), [article](#).

Vision–Language Models: why pair vision with language?

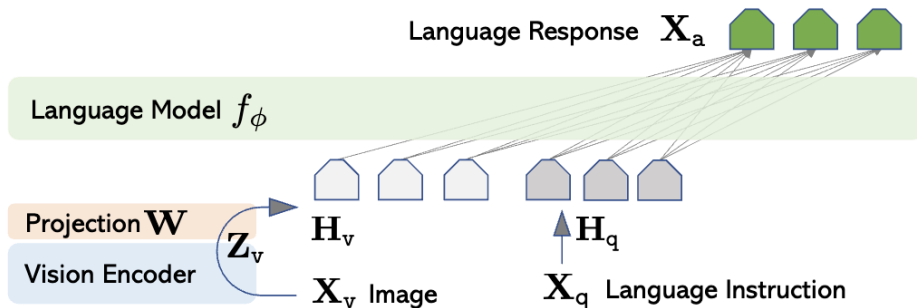
- ▶ Strong **image embeddings** are useful, but many tasks still need a way to **say what you want** in words (“what species of bird is this?”, “read the sign in the corner”).
- ▶ **Language** makes it easy to specify new tasks without collecting a fresh classification label set for every concept—a path to flexible, user-steerable vision AI.
- ▶ **Nomenclature (informal)**: an **LLM** is language-first; a **VLM** adds a **vision encoder** so the stack can **condition on images**; an **MLLM** usually means several modalities (image, video, audio) in one family.

Bandaru, *Vision Language Models* (blog, Aug. 2025), rohitbandaru.github.io/blog/Vision-Language-Models/.

Fusion Patterns: Where do image tokens meet the LLM?

- ▶ Early fusion: build one long token sequence (visual tokens + text tokens) and run attention over the mixture—what most “native” VLMs aim for today.
- ▶ Middle fusion: insert vision cross-attention inside some transformer blocks (common when adapting a legacy text-only stack such as early LLaMA 3.2).
- ▶ Late fusion: combine modalities only at the end—usually too weak for open-ended captioning/VQA because the LLM never gets rich visual state early enough.
- ▶ Images create many tokens; good designs compress visual length (pooling, grouping, tiling) before the LLM sees them.

- ▶ Freeze (or lightly train) a strong ViT; map its patch tokens through a small MLP projector into the LLM embedding width, then concatenate with text tokens.
- ▶ Simple, stable, and very common in open recipes (LLaVA, LLaMA 4-style early fusion, PaliGemma 2, DeepSeek-VL, ...).



Diagram

from Bandaru. Paper: Liu et al., LLaVA, [arXiv:2304.08485](https://arxiv.org/abs/2304.08485).

- ▶ A small **adapter** uses learned query vectors that **cross-attend** to ViT tokens and output a **fixed small** set of visual summaries for the LLM.

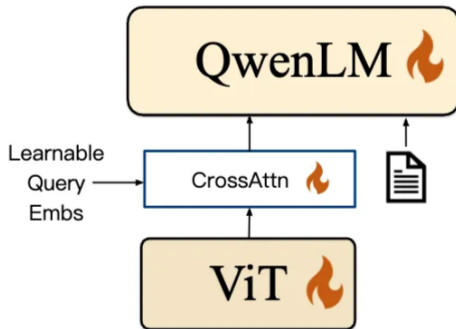


Diagram from Bandaru (Qwen-VL "position-aware" adapter).

Middle Fusion Approaches: Flamingo

- Keep a pretrained ViT and LLM frozen; insert Perceiver resamplers (compress variable tokens → fixed count) and gated cross-attention blocks inside the LLM stack.

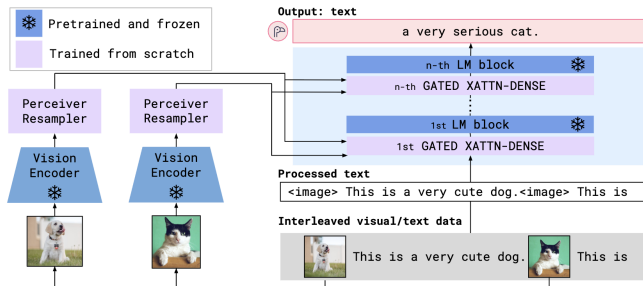


Diagram from Bandaru. Paper: Alayrac et al., Flamingo, [arXiv:2204.14198](https://arxiv.org/abs/2204.14198).

- ▶ **ViT encoder:** 32-layer ViT trained CLIP-style for image-text alignment; features extracted from multiple layers.
- ▶ **Decoder:** Image cross attention in every fourth block (no gating on cross attention).
- ▶ **Mid fusion** suits LLaMA 3's text-first design; LLaMA 4 moves to early fusion (MLP adapter) for true multimodal training.

Summary: Transformers 2017 vs. 2026

Component	2017 Transformer	2026 consensus
Attention mechanism	Multi-Head Attention (MHA)	Grouped-Query Attention (GQA) / Multi-Head Latent Attention (MLA)
Position encoding	Absolute sinusoidal (or learned absolute)	Rotary Positional Embeddings (RoPE)
Normalization	Post-Layer Normalization (post-LN)	Pre-Layer Normalization with RMSNorm
Activation	ReLU in the FFN	SwiGLU / gated linear units
Bias terms	Biases in linear layers common	Bias-free or minimal-bias architectures
Parameter density	Dense (all parameters active per forward)	Sparse activation (e.g., Mixture-of-Experts)

- ▶ The slowest part is often getting data in and out of memory, not just the math.
- ▶ New models (2024–2026) are designed together, optimizing attention, normalization, position encoding, sparsity (MoE), and serving (memory, paging, kernels).
- ▶ The main goals: cut down on moving data, adjust compute to the input's length and type, and add more parameters without needing a lot more computation per token.

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